

Generalized linear models

Models for binary outcome

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Models for binary outcome - Contents

Models for binary outcome / response

Binary outcome

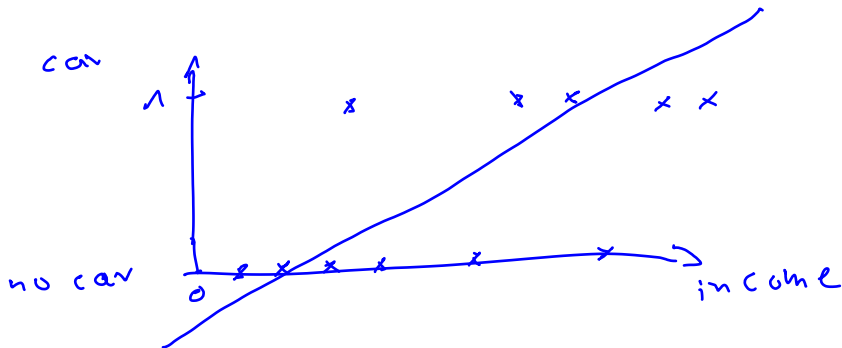
$$P(y=1 | \underline{x}) = \pi(\underline{x})$$

$$\Rightarrow P(y=0 | \underline{x}) = 1 - \pi(\underline{x})$$

Let $y \in \{0, 1\}$ be a binary response variable. Is

$$\pi = \pi(\underline{x}) = \mathbb{E}(y | \mathbf{x}) = \beta_0 + \mathbf{x}^T \beta \in [0, 1] \quad \leftarrow \text{no sense!}$$

reasonable?



Binary outcome

Reasons that speak against this model approach:

- range of reasonable values for our model doesn't fit $\rightarrow$ could be > 1 or < 0
- Variance homogeneity

$$y \sim \text{Ber}(\pi) \Rightarrow \text{Var}(y) = \pi(1-\pi)$$

$$\pi(x_i) \rightarrow \pi(x_i)(1-\pi(x_i))$$

not constant

Motivating the binary model via Bayes' theorem

Let $(y, \mathbf{x})^\top$ be a random vector with

$$y \in \{0, 1\}, \quad \mathbf{x} = (x_1, \dots, x_p)^\top$$

and $\mathbf{x} \mid y = i \sim N(\mu_i, \Sigma)$.

(this assumption only makes sense for metric covariates)

Recall: Bayes' theorem

$$P(y \mid x) = \frac{P(x \mid y)P(y)}{P(x)}$$

Motivating the binary model via Bayes' theorem

For continuous x and binary y :
 $\pi(\underline{x})$ " "
 $\mathbb{E}[\gamma | \underline{x}]$ " "
cond. density *prior prob.*

$$P(y = 1 | \underline{x}) = \frac{f(\underline{x}|y=1)P(y=1)}{f(\underline{x})} =$$

marginal density

$$= \frac{f(\underline{x}|y=1)P(y=1)}{f(\underline{x}|y=1)P(y=1) + f(\underline{x}|y=0)P(y=0)}$$

$$= \frac{\frac{f(\underline{x}|y=1)P(y=1)}{f(\underline{x}|y=0)P(y=0)}}{\frac{f(\underline{x}|y=1)P(y=1)}{f(\underline{x}|y=0)P(y=0)} + 1}$$

Motivating the binary model via Bayes' theorem

Taking the logarithm of the numerator provides

$$\begin{aligned}
 & \log \left(\frac{P(\underline{x} | y=1) P(y=1)}{P(\underline{x} | y=0) P(y=0)} \right) = \\
 & \log \left(\frac{\frac{1}{(2\pi)^{p/2}} |\Sigma^{-1}|^{-1/2} \exp\left(-\frac{1}{2} (\underline{x} - \underline{\mu}_1)^T \Sigma^{-1} (\underline{x} - \underline{\mu}_1)\right) P(y=1)}{\frac{1}{(2\pi)^{p/2}} |\Sigma^{-1}|^{-1/2} \exp\left(-\frac{1}{2} (\underline{x} - \underline{\mu}_0)^T \Sigma^{-1} (\underline{x} - \underline{\mu}_0)\right) P(y=0)} \right) = \\
 & = -\frac{1}{2} \left[\cancel{\underline{x}^T \Sigma^{-1} \underline{x}} - 2 \underline{\mu}_1^T \Sigma^{-1} \underline{x} + \underline{\mu}_1^T \Sigma^{-1} \underline{\mu}_1 - \cancel{\underline{x}^T \Sigma^{-1} \underline{x}} \right. \\
 & \quad \left. + 2 \underline{\mu}_0^T \Sigma^{-1} \underline{x} - \underline{\mu}_0^T \Sigma^{-1} \underline{\mu}_0 \right] + \log \left(\frac{P(y=1)}{P(y=0)} \right) \\
 & = \underbrace{-\frac{1}{2} (\underline{\mu}_1^T \Sigma^{-1} \underline{\mu}_1 - \underline{\mu}_0^T \Sigma^{-1} \underline{\mu}_0)}_{=: \beta_0} + \log \left(\frac{P(y=1)}{P(y=0)} \right) + \underbrace{\underline{x}^T \Sigma^{-1} (\underline{\mu}_1 - \underline{\mu}_0)}_{=: \underline{x}^T \underline{\beta}}
 \end{aligned}$$

Motivating the binary model via Bayes' theorem

Plugging this into the fraction

$$\log(\dots) = \beta_0 + \underline{x}^T \underline{\beta} \Rightarrow (\dots) = \exp(\beta_0 + \underline{x}^T \underline{\beta})$$

$$\Rightarrow P(Y=1 | \underline{x}) = \frac{\exp(\beta_0 + \underline{x}^T \underline{\beta})}{\exp(\beta_0 + \underline{x}^T \underline{\beta}) + 1}$$

$$\Pi(\underline{x}) = \overline{F}(\beta_0 + \underline{x}^T \underline{\beta})$$

$$\overline{F}(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)} \quad \text{logistic (distribution) function}$$

$$\text{More general: } P(Y=1 | \underline{x}) = \text{Transf. of } \beta_0 + \underline{x}^T \underline{\beta}$$

Motivating the binary model via latent variables

Basic idea:

- ▶ We observe a binary response $y \in \{0, 1\}$
- ▶ However: we assume underlying continuous variable
- Not observable or unknown: **latent**
- ▶ actually continuous $\tilde{y}$ in the background

Motivating the binary model via latent variables

Example

Let

- ▶ y_i : response variable
- ▶ x_i : covariate
- ▶ $\tilde{y}_i$: latent metric variable

For example:

- ▶ y_i : death yes/no
- ▶ x_i : dosis of poison
- ▶ $\tilde{y}_i$: health damage

Or:

- ▶ y_i : car yes/no
- ▶ x_i : income
- ▶ $\tilde{y}_i$: welfare of the household

Motivating the binary model via latent variables

Assumption

$\tilde{y}_i$ follows the classical linear model

$$\tilde{y}_i = \gamma_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i,$$

where $-\varepsilon_i$ has the distribution function F .

$$\begin{aligned} -\varepsilon \sim F &\Rightarrow \mathbb{P}(-\varepsilon \leq x) = \mathbb{P}(\varepsilon \geq -x) = F(x) \\ \text{and also} \\ \Rightarrow \mathbb{P}(\varepsilon \leq x) &= 1 - \mathbb{P}(\varepsilon \geq x) = 1 - F(-x) \end{aligned}$$

Motivating the binary model via latent variables

Relationship between y and $\tilde{y}$

Observation and latent variable

$$y_i = 1 \quad \Leftrightarrow \quad \tilde{y}_i \geq \vartheta$$

 threshold

For example:

death occurs $\Leftrightarrow$ toxicity threshold exceeds

Motivating the binary model via latent variables

Then we obtain:

$$\begin{aligned} \pi_i(\underline{x}_i) & \stackrel{!}{=} P(y_i = 1 \mid \underline{x}_i) = P(\tilde{y}_i \geq \nu \mid \underline{x}_i) \\ & = P(\gamma_0 + \underline{x}_i^T \beta + \varepsilon_i \geq \nu \mid \underline{x}_i) \\ & = P(\underbrace{\gamma_0 + \underline{x}_i^T \beta - \nu}_{\underbrace{\gamma_0 - \nu + \underline{x}_i^T \beta}_{=:\beta_0}} \geq \underbrace{-\varepsilon_i}_{\sim F} \mid \underline{x}_i) \\ & = F_{-\varepsilon_i}(\beta_0 + \underline{x}_i^T \beta) \end{aligned}$$

Motivating the binary model via latent variables

Let

$$\pi_i(\mathbf{x}_i) = P(y_i = 1 \mid \mathbf{x}_i) = F(\beta_0 + \mathbf{x}_i^\top \beta)$$

Depending on the choice of $F(\cdot)$ different models result:

1. $F(\cdot) = \text{logistic}$: $F(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)}$ logit-Model
most frequently used
2. $F(\cdot) = \Phi$ Gaussian cdf $\Rightarrow$ probit-Model
only rarely used
3. $F(\cdot) = \text{Gumbel, Gompertz, skewed distributions,}$
only very rarely used